UAVEDUSim: An Educational Simulator of Jamming Attacks in UAVs

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***Abstract*—Unmanned Aerial Vehicles (UAVs) are widely used in civilian, industrial, and defense applications, where Global Navigation Satellite System (GNSS) signals are a primary source of positioning and navigation. Interference with these signals, known as jamming, poses a growing threat to UAV navigation and communication. However, regulatory restrictions, specialized hardware requirements, and the risk of interfering with nearby systems limit real-world evaluation, making simulation an important alternative for research and education. This paper presents UAVEDUSim, an educational simulator based on PX4 SITL and modern Gazebo, featuring a guided graphical interface for configuring GNSS jamming attacks (Noise, Blackout, and Pulse). The work also presents a preliminary evaluation of quiz-based gamification as a pedagogical approach for teaching the effects of different jamming patterns on UAV behavior. An evaluation with 10 participants indicated high usability, with all users reporting that the installation process was straightforward. The simulator received high satisfaction (8.75-9.0/10) and perceived usefulness ratings (9.75-9.83/10). These results indicate that UAVEDUSim was perceived as an accessible and useful platform for teaching UAV jamming concepts through interactive simulation and gamification.**

***Index Terms*—Unmanned Aerial Vehicles, Jamming Attacks, Simulation, Gamification, Cybersecurity.**

I. Introduction

The use of Unmanned Aerial Vehicles (UAVs) has expanded significantly across civilian, industrial, and defense sectors, driven by cost reductions, advances in onboard sensors, and gains in operational autonomy [1], [2]. However, these systems rely on radio links and satellite signals to determine their position during flight [1], [3], making any degradation, blocking, or manipulation of those signals a direct risk to operational safety.

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Among the threats that exploit this reliance, jamming stands out for its ability to compromise a UAV's communication and navigation links using relatively simple hardware, requiring only the transmission of radio-frequency (RF) interference over the same spectrum band used by legitimate systems [3]–[6]. When the target is the Global Positioning System (GPS) or other Global Navigation Satellite System (GNSS) systems, the attack disrupts positioning measurements or navigation communication directly [3], [7], [8], which can result in loss of control, route deviation, or the triggering of emergency landings in unsuitable locations [3]. In this context, it is fundamental to practice how each attack pattern alters UAV behavior, and how countermeasures reverse that effect, a skill that cybersecurity professionals and students, as well as UAV operators, need to develop through hands-on training, not only through theoretical study.

Under real-world conditions, conducting jamming attacks is illegal due to regulatory and operational restrictions, as such attacks can affect other critical systems. In general, transmitting intentional interference requires controlled environments that often fail to reflect realistic usage scenarios [9]. Given these constraints, simulation environments are an important alternative not only for research on jamming attacks but also as a teaching tool that enables this knowledge to be taught in a controlled, safe, and reproducible setting [10], [11].

In recent years, researchers have proposed educational approaches involving UAVs, ranging from simulators and virtual environments to structured hands-on exercises. Virtual training systems have been developed to teach autonomous and teleoperated UAV control, reproducing realistic scenarios and enabling validation of control algorithms without the cost and risk associated with physical equipment [12]. In ecological education contexts, UAVs have been used as a hands-on tool to teach field data acquisition, with students organized into groups and the flight area zoned to ensure safety and data coverage [13]. In the cybersecurity domain,

UCLP [14] offers a hands-on laboratory platform, built on PX4 and Gazebo, with pre-configured attack and defense modules, including a GPS jamming module, for educational purposes, while the ProVANT Simulator offers a simulation environment for developing and validating control strategies, without an educational purpose [15].

Although these works demonstrate the value of simulation and hands-on practice across several educational domains, none specifically focuses on teaching how different jamming attack patterns affect UAV behavior and how mitigation strategies affect that behavior. This understanding is far from trivial: a full signal blackout, in which the receiver stops obtaining position updates altogether, and a degradation caused by continuous noise, in which readings keep arriving but with reduced accuracy, produce quite distinct effects on UAV navigation, and this difference is difficult to convey through theoretical lectures or technical papers alone. In recent years, gamification has proven an effective strategy for supporting this kind of experiential learning [16]. None of the works discussed above, however, evaluates gamification specifically as a strategy for teaching different jamming attack patterns and their mitigations. This work therefore assesses gamification, through mechanisms such as knowledge-check quizzes integrated directly into the simulator's flow, as a pedagogical tool for this domain.

To address this gap, this work presents UAVEDUSim, an educational, interactive simulator with a graphical interface built on PX4 SITL and modern Gazebo, whose central pedagogical focus is teaching jamming attacks on UAVs and their respective mitigations. UAVEDUSim combines GPS/GNSS attack and countermeasure simulation with a guided graphical interface that provides scenario selection, interactive attack control, tutorials, and a quiz-based learning assessment mechanism, allowing students without advanced RF or offensive security knowledge to safely and reproducibly explore the effects of jamming on UAV navigation.

The main contributions of this work are summarized as follows:

- UAVEDUSim, an educational simulator with a focus on teaching jamming attacks and mitigations on UAVs;
- The development of a graphical interface that guides the user through scenario selection, attack execution, and visualization of its effects and countermeasures;
- A quiz-based gamification mechanism that checks the user's understanding of the presented concepts and requires a minimum performance level (2 out of 5 correct answers) to unlock access to the simulation.

This work is organized as follows: Section 2 presents the related work. Section 3 details the methodology of this work. Section 4 presents the results, findings, and limitations. Finally, Section 5 presents the conclusions and directions for future work.

II. Related Work

Several studies have explored simulators, virtual environments, and hands-on practice with real UAVs for teaching these systems, each addressing a distinct objective. A virtual training system based on Unity3D and MATLAB was developed to teach autonomous and teleoperated UAV control, including the kinematic and dynamic modeling of a real hexacopter [12]. Although the system teaches control algorithms and received good usability scores, it did not model cyberattacks or jamming.

Li *et al.* (2026) [13] proposed an experimental teaching case in which students use real UAVs for field data acquisition, organized into groups with area zoning to ensure safety and flight coverage. In this context, Félix-Herrán *et al.* (2022) [17] presented an intensive Challenge-Based Learning (CBL) course in which engineering students program autonomous routines on real UAVs to develop disciplinary and transversal competencies. Somerville *et al.* (2024) [18] developed a controlled experimental design that showed that pre-training on a commercial simulator significantly reduces pilots' positioning error during subsequent real-world flights. These three works reinforce the pedagogical value of simulator environments for UAVs in developing field sensing, autonomous flight programming, and manual piloting, but none of them address jamming attacks or their effects on navigation.

Closer to the security domain, a gamified generic drone simulator was developed for search-and-rescue (SAR) mission training, balancing interface design, physical accuracy, and environmental realism, and validated with 57 participants, including pilots, engineers, and air traffic controllers [16]. This work highlights gamification's potential to foster motivation and learning retention in UAV simulators, but its pedagogical purpose is oriented entirely toward search-and-rescue operations, with no modeling of cyberattacks.

Regarding high-fidelity UAV simulators, the ProVANT Simulator offers a robust environment, built on Gazebo and ROS, for developing and validating control algorithms across different UAV configurations, including quadrotors, tilt-rotors, and aerial manipulators, with support for Software-in-the-Loop and Hardware-in-the-Loop testing [15]. Despite its technical robustness and the compatibility of its technological base (Gazebo) with that adopted in UAVEDUSim, ProVANT is oriented toward research and industrial development, not education, and does not address cybersecurity.

Finally, the work most closely related to the scope of this research is the UAV Cybersecurity Laboratory Platform (UCLP), which proposes a hands-on laboratory platform, built on PX4 and Gazebo, with pre-configured attack and defense modules organized in a plug-in design, specifically aimed at training UAV cybersecurity professionals [14]. Among the reviewed works, UCLP is the only one that models a GPS jamming attack in UAV cybersecurity. However, the UCLP jamming model is limited to complete GPS signal interruption and does not account for variations such as continuous noise or intermittent interference.

Table I summarizes the characteristics of these works compared to UAVEDUSim. Taken together, the reviewed works show that simulation and hands-on practice with real hardware are already widely recognized as effective educational tools for

UAVs. However, none combine specific modeling of multiple GPS/GNSS jamming patterns with a guided graphical interface and a gamification-based learning-assessment mechanism, which justifies the UAVEDUSim contributions proposed in this work.

III. Methodology

A. System Architecture Overview

UAVEDUSim integrates three commonly used components for UAV simulation: the PX4 Autopilot operating in Software-in-the-Loop (SITL) mode, the modern Gazebo physics engine, and QGroundControl as the ground control station. These components communicate through the Micro Air Vehicle Link (MAVLink) protocol. UAVEDUSim's educational layer is implemented through a Python Tkinter graphical user interface. This interface orchestrates simulation initialization and termination, guides users through scenario configuration, and centralizes interactive control of jamming attacks and their corresponding mitigation strategies. As a result, users do not need to interact directly with PX4 or Gazebo terminals or configuration files. Figure 1 shows the four components running simultaneously during a simulation session.

During execution, jamming attacks are injected directly into the GPS sensor data PX4 processes. Noise jamming perturbs position, altitude, velocity, and GPS quality indicators; Blackout interrupts GPS updates; and Pulse alternates between degraded, unavailable, recovering, and valid measurements. For Noise, mitigation combines Kalman filtering, consistency checks, and auxiliary sensor information to reject or reduce corrupted GPS measurements; for Blackout, pseudo-measurements are generated using auxiliary localization information; and for Pulse, both strategies are combined according to the current interference state. Under persistent degradation or signal loss, the system uses a controlled landing procedure. The mitigation mechanisms are implemented at the source-code level, allowing existing strategies to be modified and new ones to be added, while the graphical interface provides access to the currently implemented mechanisms.

B. Educational Workflow and Gamification

Before accessing the simulation, users follow a guided educational workflow. When the application starts, users are prompted to select the interface language (Portuguese or English), which is then applied to all textual elements of the simulator. The home screen is subsequently displayed with the "Start Simulation" option initially disabled, along with a navigation menu that provides access to educational content on jamming concepts, mitigation strategies, and the knowledge assessment quiz. The educational screens are organized into a guided sequence, with each step leading to the next topic, although the navigation menu also allows direct access to any educational screen or the quiz at any time.

Under the standard workflow, simulation access is granted only after completing a five-question multiple-choice quiz covering the presented concepts, with at least two correct answers required to unlock the "Start Simulation" option. During the quiz, users receive immediate feedback on each answer, including the correct option and an additional explanation when they submit an incorrect answer. After users answer all five questions, the final score is displayed. If the minimum passing score is not achieved, users may restart the quiz without restrictions. For each new attempt, the order of the answer choices is randomized to discourage memorizing the positions of the answer choices rather than the underlying concepts. Once the quiz is successfully completed, the unlocked state is preserved across future application executions, eliminating the need to repeat the educational process. For comparison during the user evaluation, we enabled direct simulation access for four participants, while six followed the standard workflow.

Figures 2 and 3 show examples of the interfaces associated with this stage. This gamification mechanism is intended to promote assimilation of jamming and mitigation concepts before users interact with the simulation, structuring the learning process rather than merely providing a simulation environment without prior assessment.

C. Scenario Configuration and Attack Control

After the simulation is unlocked, users configure the scenario in two stages. First, users select one of 15 UAV models, including the recommended `gz_x500_uavjamsim` model, equipped with multiple onboard sensors, and a lightweight alternative that also supports visual odometry for mitigation strategies while requiring fewer computational resources. Second, one of six Gazebo environments is selected, including an empty world recommended for initial experiments and a world containing water and vegetation, which is useful for observing emergency landings triggered by the mitigation system over unsuitable terrain. Before executing any attack, the simulator provides an integrated tutorial that describes the platform's operation and how to configure jamming attack parameters, including how intensity controls the magnitude of GPS disturbances in the Noise attack and the duration of signal interruption in the Pulse attack, while the Blackout attack has no intensity parameter.

Once the scenario is configured, users gain access to the attack control panel, which remains disabled until the takeoff command is issued and confirmed by PX4. After takeoff confirmation, waypoint transmission remains temporarily unavailable for a configurable delay (12 s by default), displayed through a countdown timer to ensure that the UAV reaches a sufficient altitude before initiating horizontal flight. Once the destination waypoint has been transmitted (distance, heading, and relative altitude are all user-configurable), the panel enables selection of the desired jamming attack (Noise, Blackout, or Pulse), activation or deactivation of the corresponding mitigation before or during the attack, and real-time monitoring of simulation events through a status log. This interface allows users to compare UAV behavior with and without mitigation under identical experimental conditions, without requiring manual commands or direct modifications to PX4 or Gazebo configuration files.

TABLE I: Comparison of educational approaches for UAVs.

Work	Educational Context	GNSS Jamming	Guided Interface	Gamification
UAV Virtual Training System [12]	Control	×	✓	×
UAV Field-Data Teaching Case [13]	Ecology	×	×	×
Drone Rally I-week [17]	Engineering	×	×	×
DJI Flight Simulator [18]	Piloting	×	×	×
Generic Drone Simulator (GDS) [16]	SAR	×	✓	✓
ProVANT Simulator [15]	Industry	×	✓	×
UCLP [14]	Cybersecurity	✓	×	×
UAVEDUSim (This work)	GPS & Jamming	✓	✓	✓

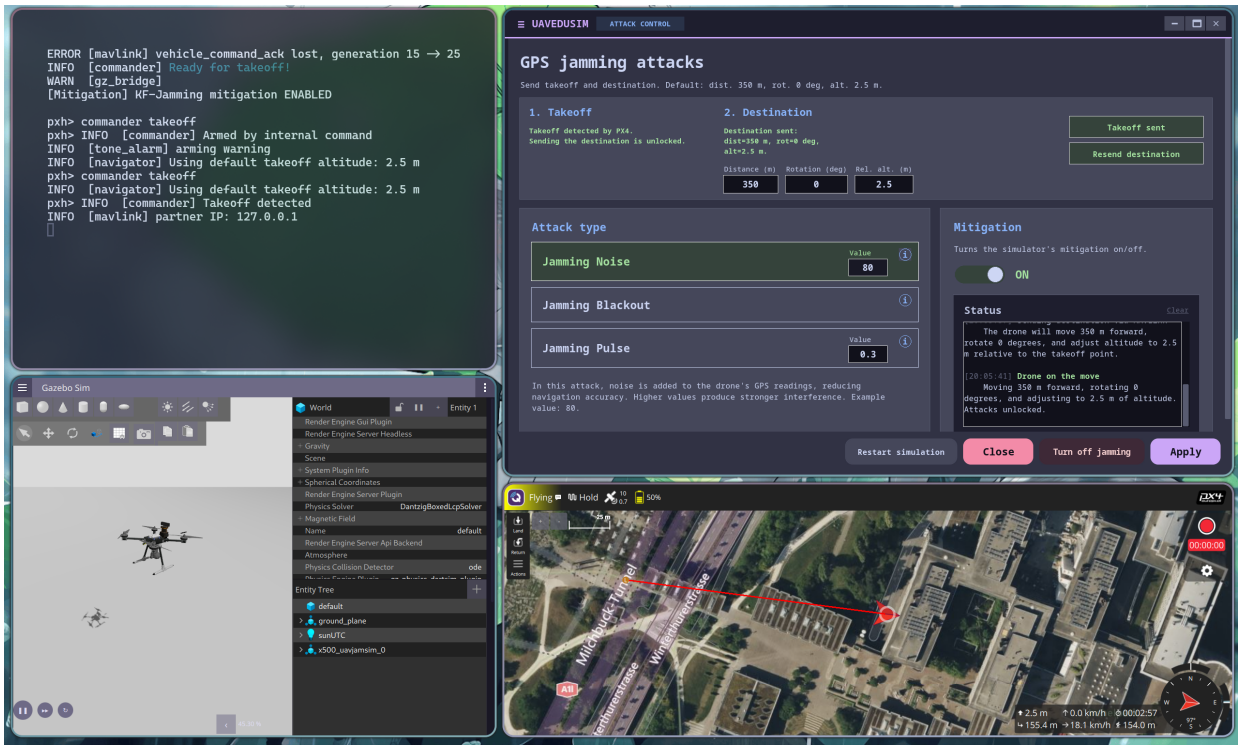


Fig. 1: UAVEDUSim components running simultaneously: the graphical user interface, the PX4 SITL terminal, the Gazebo simulation environment, and QGroundControl.

IV. Results

UAVEDUSim underwent a preliminary usability and educational evaluation with ten volunteers with different levels of familiarity with Linux systems and UAV cybersecurity concepts. Before using the simulator, 70% of participants reported familiarity with *jamming*, whereas 30% had only superficial knowledge. All participants (100%) considered the installation and execution process easy, indicating good platform usability.

Of the ten participants, six (60%) followed the standard workflow and completed the quiz before starting the simulation, while four (40%) received direct access for comparison. Table II compares the two groups regarding overall satisfaction and perceived usefulness of the simulator. Although both groups rated the educational usefulness similarly, the group that completed the quiz reported slightly higher overall satisfaction. Among participants who accessed the simulation directly, without completing the quiz beforehand, only 25% (1 out of 4) reported understanding the jamming and mitigation concepts without any additional assistance, while the remaining 75% (3 out of 4) needed to consult the simulator's explanatory screens during execution to follow what was happening.

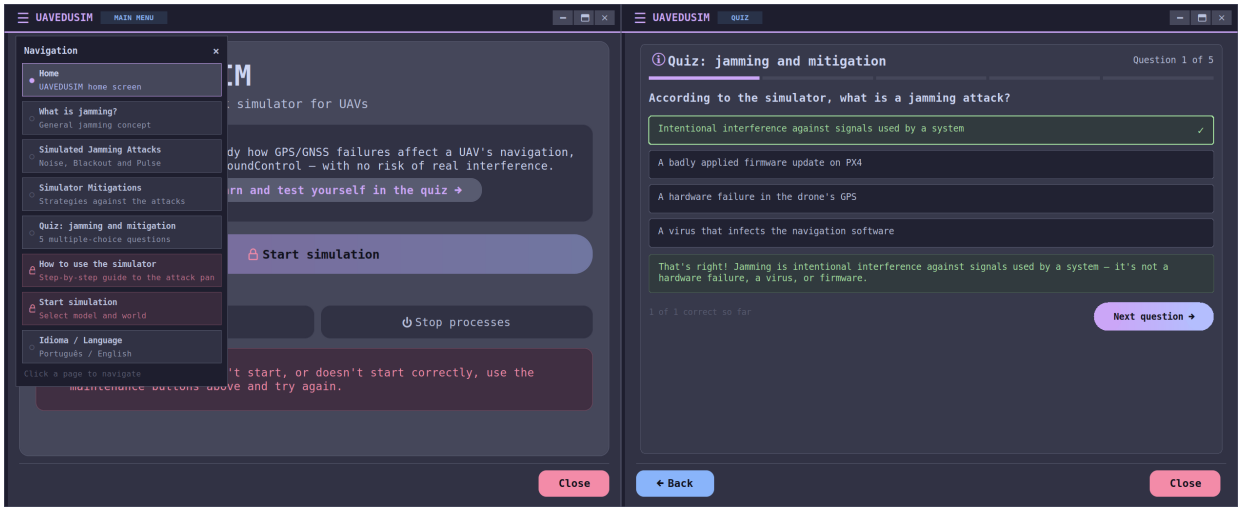


Fig. 2: Navigation menu displayed with the *Start Simulation* option disabled (left) and an example of a correctly answered quiz question with immediate feedback (right).

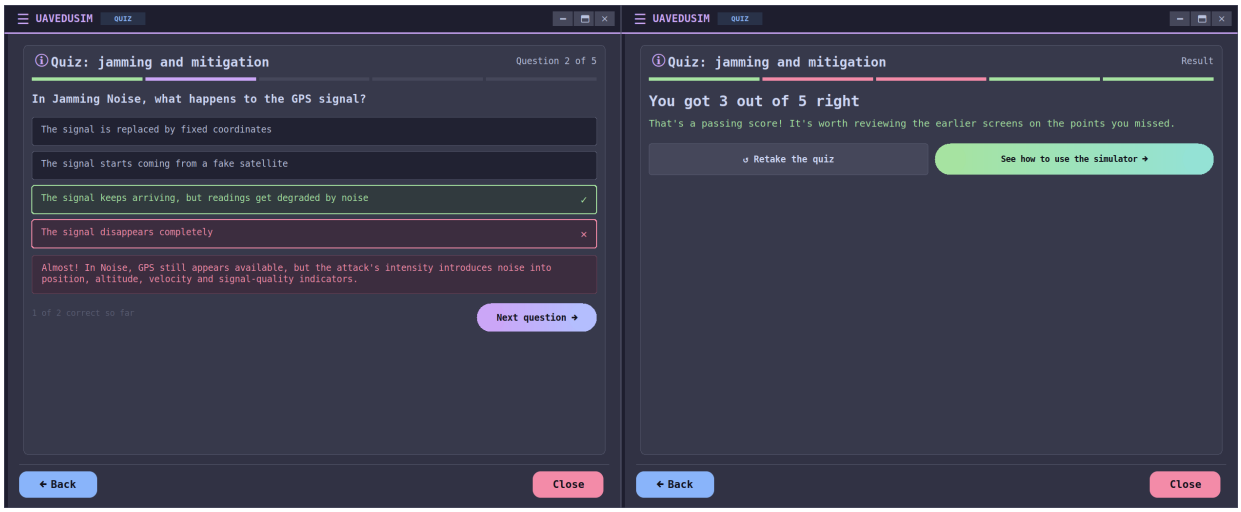


Fig. 3: Example of an incorrectly answered question showing the selected option and the correct answer (left) and the quiz results screen showing the number of correct answers and unlocking the simulation when the minimum passing criterion is met (right).

TABLE II: Comparison between participants who completed the quiz before the simulation and those who accessed it directly.

Metric	Completed the quiz	Direct access
Participants	60% (6 of 10)	40% (4 of 10)
Average overall satisfaction rating	9.0 / 10	8.75 / 10
Average perceived usefulness rating	9.83 / 10	9.75 / 10

Table III presents results of the gamification mechanism, based only on participants who completed the quiz. All of them reported that the quiz helped reinforce their understanding of the concepts presented, with an average rating of 9.5/10 for its contribution to learning. The quiz difficulty received an average rating of 4.7/10, indicating that most users found the questions accessible.

TABLE III: Evaluation of the gamification mechanism among participants who completed the quiz.

Metric	Result
Reported that the quiz helped their learning	100%
Average rating of the quiz's contribution to learning	9.5 / 10
Average quiz difficulty rating	4.7 / 10

Table IV presents the overall simulator evaluation for all ten participants. Half of the participants (50%) rated the interface as very intuitive, 40% considered it easy to use with minor room for improvement, and only one participant (10%) reported difficulties during use. The simulator received an

average satisfaction rating of 8.9/10 and a 9.8/10 rating for its usefulness as a teaching tool for UAV jamming attacks, indicating a positive result.

TABLE IV: Overall evaluation of UAVEDUSim (all participants).

Metric	Result
Installation considered easy	100%
Interface rated as very intuitive	50%
Interface rated as easy to use	40%
Participants reporting difficulties	10%
Average overall satisfaction rating	8.9 / 10
Average rating of usefulness for teaching	9.8 / 10

Although the results indicate that UAVEDUSim has potential as an educational tool, the gamification mechanism still relies on a limited set of fixed questions, which may encourage memorization of answers after repeated attempts. In addition, we conducted the evaluation with a small group of 10 volunteers and should therefore consider it preliminary, as student availability for the testing stage was limited.

As future work, we will expand the question bank to allow customizable questions with different difficulty levels and incorporate a user authentication system to enable individual progress tracking, rankings, scoring, and achievement mechanisms. A new evaluation round with more participants is also planned, using complete classes in a controlled laboratory setting to enable more statistically robust analysis.

V. Conclusion

This work presented UAVEDUSim, an educational, interactive simulator for teaching GPS/GNSS jamming attacks and their respective mitigations on UAVs, built on top of PX4 SITL and modern Gazebo. The simulator combines a guided graphical interface, which centralizes scenario selection and interactive control of the attacks (Noise, Blackout, and Pulse) and their corresponding mitigations, with a quiz-based gamification mechanism that assesses the user's understanding of the concepts presented before granting access to the simulation.

The preliminary evaluation conducted with ten volunteers indicates positive user perception of gamification as a pedagogical strategy for teaching experiential concepts that are difficult to convey through theory alone, such as UAV behavior under different attack patterns. As future work, we intend to expand the gamification mechanisms and conduct evaluations with a larger number of participants, allowing a deeper investigation of UAVEDUSim's impact on teaching UAV cybersecurity.

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